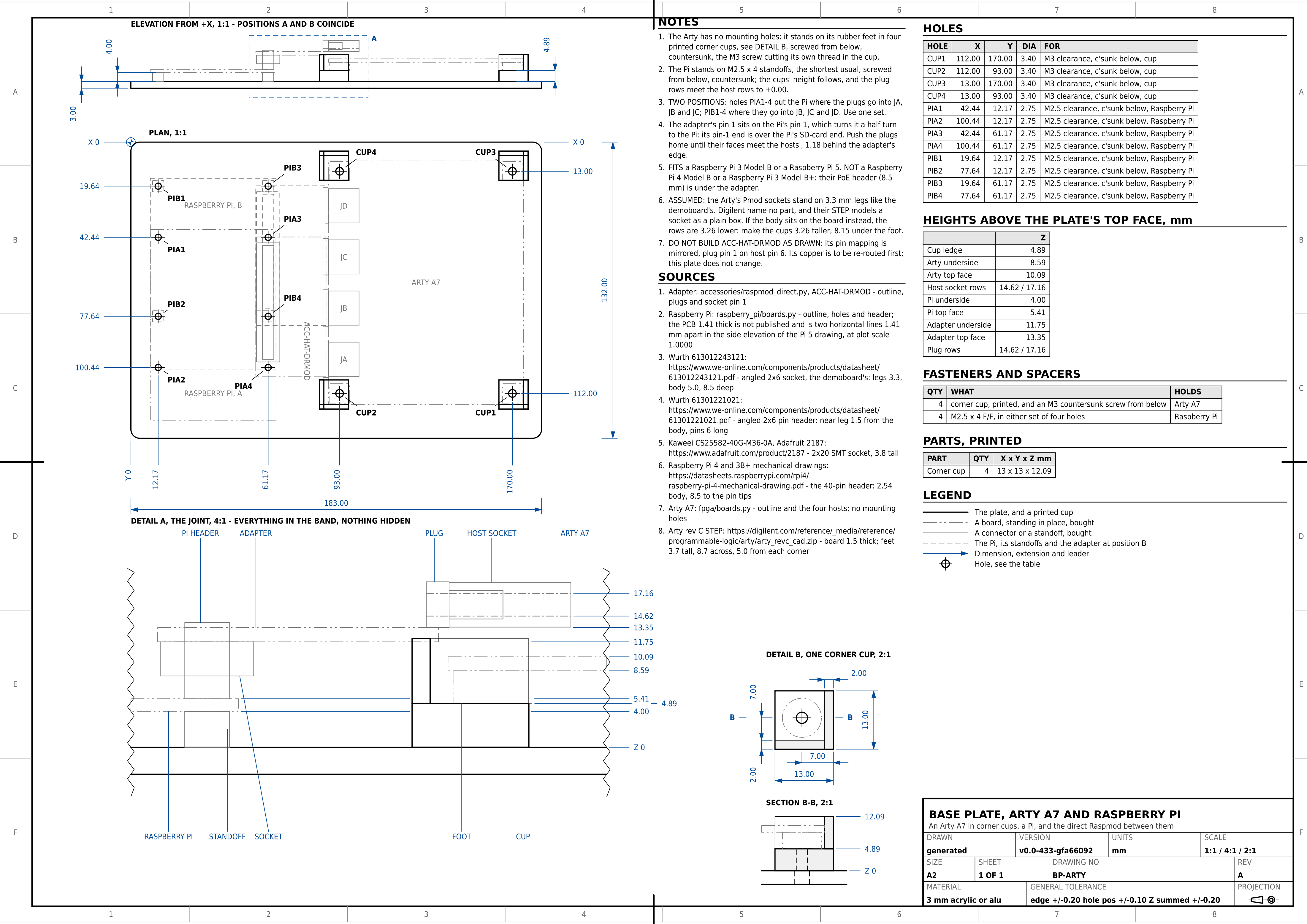




NOTES

- The Arty has no mounting holes: it stands on its rubber feet in four printed corner cups, see DETAIL B, screwed from below, countersunk, the M3 screw cutting its own thread in the cup.
- The Pi stands on M2.5 x 4 standoffs, the shortest usual, screwed from below, countersunk; the cups' height follows, and the plug rows meet the host rows to +0.00.
- TWO POSITIONS: holes PIA1-4 put the Pi where the plugs go into JA, JB and JC; PIB1-4 where they go into JB, JC and JD. Use one set.
- The adapter's pin 1 sits on the Pi's pin 1, which turns it a half turn to the Pi: its pin-1 end is over the Pi's SD-card end. Push the plugs home until their faces meet the hosts', 1.18 behind the adapter's edge.
- FITS a Raspberry Pi 3 Model B or a Raspberry Pi 5, NOT a Raspberry Pi 4 Model B or a Raspberry Pi 3 Model B+: their PoE header (8.5 mm) is under the adapter.
- ASSUMED: the Arty's Pmod sockets stand on 3.3 mm legs like the demoboard's. Diligent name no part, and their STEP models a socket as a plain box. If the body sits on the board instead, the rows are 3.26 lower: make the cups 3.26 taller, 8.15 under the foot.
- DO NOT BUILD ACC-HAT-DRMOD AS DRAWN: its pin mapping is mirrored, plug pin 1 on host pin 6. Its copper is to be re-routed first; this plate does not change.

SOURCES

- Adapter: accessories/raspmod_direct.py, ACC-HAT-DRMOD - outline, plugs and socket pin 1
- Raspberry Pi: raspberry_pi/boards.py - outline, holes and header; the PCB 1.41 thick is not published and is two horizontal lines 1.41 mm apart in the side elevation of the Pi 5 drawing, at plot scale 1.0000
- Würth 613012243121: <https://www.we-online.com/components/products/datasheet/613012243121.pdf> - angled 2x6 socket, the demoboard's: legs 3.3, body 5.0, 8.5 deep
- Würth 61301221021: <https://www.we-online.com/components/products/datasheet/61301221021.pdf> - angled 2x6 pin header: near leg 1.5 from the body, pins 6 long
- Kaweei CS25582-40G-M36-0A, Adafruit 2187: <https://www.adafruit.com/product/2187> - 2x20 SMT socket, 3.8 tall
- Raspberry Pi 4 and 3B+ mechanical drawings: <https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-mechanical-drawing.pdf> - the 40-pin header: 2.54 body, 8.5 to the pin tips
- Arty A7: fpga/boards.py - outline and the four hosts; no mounting holes
- Arty rev C STEP: https://diligent.com/reference/_media/reference/programmable-logic/arty/arty_rev_cad.zip - board 1.5 thick; feet 3.7 tall, 8.7 across, 5.0 from each corner

HOLES

HOLE	X	Y	DIA	FOR
CUP1	112.00	170.00	3.40	M3 clearance, c'sunk below, cup
CUP2	112.00	93.00	3.40	M3 clearance, c'sunk below, cup
CUP3	13.00	170.00	3.40	M3 clearance, c'sunk below, cup
CUP4	13.00	93.00	3.40	M3 clearance, c'sunk below, cup
PIA1	42.44	12.17	2.75	M2.5 clearance, c'sunk below, Raspberry Pi
PIA2	100.44	12.17	2.75	M2.5 clearance, c'sunk below, Raspberry Pi
PIA3	42.44	61.17	2.75	M2.5 clearance, c'sunk below, Raspberry Pi
PIA4	100.44	61.17	2.75	M2.5 clearance, c'sunk below, Raspberry Pi
PIB1	19.64	12.17	2.75	M2.5 clearance, c'sunk below, Raspberry Pi
PIB2	77.64	12.17	2.75	M2.5 clearance, c'sunk below, Raspberry Pi
PIB3	19.64	61.17	2.75	M2.5 clearance, c'sunk below, Raspberry Pi
PIB4	77.64	61.17	2.75	M2.5 clearance, c'sunk below, Raspberry Pi

HEIGHTS ABOVE THE PLATE'S TOP FACE, mm

	Z
Cup ledge	4.89
Arty underside	8.59
Arty top face	10.09
Host socket rows	14.62 / 17.16
Pi underside	4.00
Pi top face	5.41
Adapter underside	11.75
Adapter top face	13.35
Plug rows	14.62 / 17.16

FASTENERS AND SPACERS

QTY	WHAT	HOLDS
4	corner cup, printed, and an M3 countersunk screw from below	Arty A7
4	M2.5 x 4 F/F, in either set of four holes	Raspberry Pi

PARTS, PRINTED

PART	QTY	X x Y x Z mm
Corner cup	4	13 x 13 x 12.09

LEGEND

- The plate, and a printed cup
- A board, standing in place, bought
- A connector or a standoff, bought
- The Pi, its standoffs and the adapter at position B
- Dimension, extension and leader
- Hole, see the table

BASE PLATE, ARTY A7 AND RASPBERRY PI

An Arty A7 in corner cups, a Pi, and the direct Raspmod between them

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-433-gfa66092	mm	1:1 / 4:1 / 2:1
SIZE	SHEET	DRAWING NO	REV
A2	1 OF 1	BP-ARTY	A
MATERIAL	GENERAL TOLERANCE		PROJECTION
3 mm acrylic or alu	edge +/-0.20 hole pos +/-0.10 Z summed +/-0.20		